Theorem A9. *Starting from the sequence end at $t = T$, as long as $\omega(s_t, a_t) \geqslant \tilde{\omega}(s_t, a_t)$ holds also the following holds:*

$$V(s_t, a_t) \;\geqslant\; \widetilde{V}(s_t, a_t) \,. \tag{A216}$$

If for (s_t, a_t) the strict inequality $\omega(s_t, a_t) > \tilde{\omega}(s_t, a_t)$ holds, then we have the strict inequality

$$V(s_t, a_t) \;>\; \widetilde{V}(s_t, a_t) \,. \tag{A217}$$

If $p(s_t, a_t \mid s_{t-1}, a_{t-1}) \neq 0$ for some (s_{t-1}, a_{t-1}) then

$$\mathrm{E}_{s_t, a_t}\left[V(s_t, a_t) \mid s_{t-1}, a_{t-1}\right] \;>\; \mathrm{E}_{s_t, a_t}\left[\widetilde{V}(s_t, a_t) \mid s_{t-1}, a_{t-1}\right] \,. \tag{A218}$$

Therefore, the strict inequality $\omega(s_t, a_t) > \tilde{\omega}(s_t, a_t)$ is propagated back as a strict inequality of variances.

Proof. Proof by induction: Induction base: $V(s_{T+1}, a_{T+1}) = \widetilde{V}(s_{T+1}, a_{T+1}) = 0$ and $\omega(s_T, a_T) = \tilde{\omega}(s_T, a_T) = 0$.
Induction step ($(t+1) \to t$): The induction hypothesis is that for all (s_{t+1}, a_{t+1}) we have

$$V(s_{t+1}, a_{t+1}) \;\geqslant\; \widetilde{V}(s_{t+1}, a_{t+1}) \tag{A219}$$

and $\omega(s_t, a_t) \geqslant \tilde{\omega}(s_t, a_t)$. It follows that

$$\mathrm{E}_{s_{t+1}, a_{t+1}}\left[V(s_{t+1}, a_{t+1})\right] \;\geqslant\; \mathrm{E}_{s_{t+1}, a_{t+1}}\left[\widetilde{V}(s_{t+1}, a_{t+1})\right] \,. \tag{A220}$$

We obtain

$$\begin{aligned} & V(s_t, a_t) \;-\; \widetilde{V}(s_t, a_t) \\ &= \left(\omega(s_t, a_t) \;+\; \mathrm{E}_{s_{t+1}, a_{t+1}}\left[V(s_{t+1}, a_{t+1})\right]\right) \;-\; \left(\tilde{\omega}(s_t, a_t) \;+\; \mathrm{E}_{s_{t+1}, a_{t+1}}\left[\widetilde{V}(s_{t+1}, a_{t+1})\right]\right) \\ &= \omega(s_t, a_t) \;-\; \tilde{\omega}(s_t, a_t) \;\geqslant\; 0 \,. \end{aligned} \tag{A221}$$

If for (s_t, a_t) the strict inequality $\omega(s_t, a_t) > \tilde{\omega}(s_t, a_t)$ holds, then we have the strict inequality $V(s_t, a_t) > \widetilde{V}(s_t, a_t)$. If $p(s_t, a_t \mid s_{t-1}, a_{t-1}) \neq 0$ for some (s_{t-1}, a_{t-1}) then

$$\mathrm{E}_{s_t, a_t}\left[V(s_t, a_t) \mid s_{t-1}, a_{t-1}\right] \;>\; \mathrm{E}_{s_t, a_t}\left[\widetilde{V}(s_t, a_t) \mid s_{t-1}, a_{t-1}\right] \,. \tag{A222}$$

Therefore, the strict inequality $\omega(s_t, a_t) > \tilde{\omega}(s_t, a_t)$ is propagated back as a strict inequality of variances as long as $p(s_t, a_t \mid s_{t-1}, a_{t-1}) \neq 0$ for some (s_{t-1}, a_{t-1}).
The induction goes through as long as $\omega(s_t, a_t) \geqslant \tilde{\omega}(s_t, a_t)$. □

In Stephen Patek's PhD thesis, [89] Lemma 5.1 on page 88-89 and proof thereafter state that if $\tilde{\omega}(s, a) = \omega(s, a) - \lambda$, then the solution $\widetilde{V}^\pi$ is continuous and decreasing in λ. From the inequality above it follows that

$$\begin{aligned} & V^\pi(s, a) \;-\; \widetilde{V}^\pi(s, a) \;=\; \left(\mathrm{T}^\pi_\omega V^\pi\right)(s, a) \;-\; \left(\mathrm{T}^\pi_{\tilde{\omega}} \widetilde{V}^\pi\right)(s, a) \\ &= \omega(s, a) \;-\; \tilde{\omega}(s, a) \;+\; \mathrm{E}_{s', a'}\left[V^\pi(s', a') \;-\; \widetilde{V}^\pi(s', a') \mid s, a\right] \\ &\geqslant \omega(s, a) \;-\; \tilde{\omega}(s, a) \,. \end{aligned} \tag{A223}$$

Time-Agnostic States. We defined a Bellman operator as

$$\begin{aligned} \mathrm{T}^\pi\left[\boldsymbol{V}^\pi\right](s, a) \;&=\; \omega(s, a) \;+\; \sum_{s'} p(s' \mid s, a) \sum_{a'} \pi(a' \mid s') \, \boldsymbol{V}^\pi(s', a') \\ &=\; \omega(s, a) \;+\; \left(\boldsymbol{V}^\pi\right)^T \boldsymbol{p}(s, a) \,, \end{aligned} \tag{A224}$$

where $\boldsymbol{V}^\pi$ is the vector with value $V^\pi(s', a')$ at position (s', a') and $\boldsymbol{p}(s, a)$ is the vector with value $p(s' \mid s, a)\pi(a' \mid s')$ at position (s', a'). The fixed point equation is known as the *Bellman equation*. In vector and matrix notation the Bellman equation reads

$$\mathrm{T}^\pi\left[\boldsymbol{V}^\pi\right] \;=\; \boldsymbol{\omega} \;+\; \boldsymbol{P}\,\boldsymbol{V}^\pi \,, \tag{A225}$$

where $\boldsymbol{P}$ is the row-stochastic matrix with $p(s' \mid s,a)\pi(a' \mid s')$ at position $((s,a),(s',a'))$. We assume that the set of state-actions $\{(s,a)\}$ is equal to the set of next state-actions $\{(s',a')\}$, therefore $\boldsymbol{P}$ is a square row-stochastic matrix. This Bellman operator has the same characteristics as the Bellman operator for the action-value function q^π.
Since $\boldsymbol{P}$ is a row-stochastic matrix, the Perron-Frobenius theorem says that (1) $\boldsymbol{P}$ has as largest eigenvalue 1 for which the eigenvector corresponds to the steady state and (2) the absolute value of each (complex) eigenvalue is smaller equal 1. Only the eigenvector to eigenvalue 1 has purely positive real components. Equation 7 of Bertsekas and Tsitsiklis, 1991, [13] states that

$$\left(\mathrm{T}^\pi\right)^t \left[\boldsymbol{V}^\pi\right] \;=\; \sum_{k=0}^{t-1} \boldsymbol{P}^k\, \boldsymbol{\omega} \;+\; \boldsymbol{P}^t\, \boldsymbol{V}^\pi \;. \tag{A226}$$

Applying the operator T^π recursively t times can be written as [13]:

$$\left(\mathrm{T}^\pi\right)^t \left[\boldsymbol{V}^\pi\right] \;=\; \sum_{k=0}^{t-1} \boldsymbol{P}^k\, \boldsymbol{\omega} \;+\; \boldsymbol{P}^t\, \boldsymbol{V}^\pi \;. \tag{A227}$$

In particular for $\boldsymbol{V}^\pi = \boldsymbol{0}$, we obtain

$$\left(\mathrm{T}^\pi\right)^t \left[\boldsymbol{0}\right] \;=\; \sum_{k=0}^{t-1} \boldsymbol{P}^k\, \boldsymbol{\omega} \;. \tag{A228}$$

For finite horizon MDPs, the values $\boldsymbol{V}^\pi = \boldsymbol{0}$ are correct for time step $T+1$ since no reward for $t > T+1$ exists. Therefore, the "backward induction algorithm" [95, 96] gives the correct solution:

$$\boldsymbol{V}^\pi \;=\; \left(\mathrm{T}^\pi\right)^T \left[\boldsymbol{0}\right] \;=\; \sum_{k=0}^{T-1} \boldsymbol{P}^k\, \boldsymbol{\omega} \;. \tag{A229}$$

The product of square stochastic matrices is a stochastic matrix, therefore $\boldsymbol{P}^k$ is a stochastic matrix. Perron-Frobenius theorem states that the spectral radius $\mathcal{R}(\boldsymbol{P}^k)$ of the stochastic matrix $\boldsymbol{P}^k$ is: $\mathcal{R}(\boldsymbol{P}^k) = 1$. Furthermore, the largest eigenvalue is 1 and all eigenvalues have absolute values smaller or equal one. Therefore, $\boldsymbol{\omega}$ can have large influence on $\boldsymbol{V}^\pi$ at every time step.

Time-Aware States. Next we consider time-aware MDPs, where transitions occur only from states s_t to s_{t+1}. The transition matrix from states s_t to s_{t+1} is denoted by $\boldsymbol{P}_t$. We assume that $\boldsymbol{P}_t$ are row-stochastic matrices which are rectangular, that is $\boldsymbol{P}_t \in \mathbb{R}^{m \times n}$.

Definition A12. *A row-stochastic matrix $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ has non-negative entries and the entries of each row sum up to one.*

It is known that the product of square stochastic matrices $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ is a stochastic matrix. We show in next theorem that this holds also for rectangular matrices.

Lemma A4. *The product $\boldsymbol{C} = \boldsymbol{A}\boldsymbol{B}$ with $\boldsymbol{C} \in \mathbb{R}^{m \times k}$ of a row-stochastic matrix $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ and a row-stochastic matrix $\boldsymbol{B} \in \mathbb{R}^{n \times k}$ is row-stochastic.*

Proof. All entries of $\boldsymbol{C}$ are non-negative since they are sums and products of non-negative entries of $\boldsymbol{A}$ and $\boldsymbol{B}$. The row-entries of $\boldsymbol{C}$ sum up to one:

$$\sum_k C_{ik} \;=\; \sum_k \sum_j A_{ij}\, B_{jk} \;=\; \sum_j A_{ij} \sum_k B_{jk} \;=\; \sum_j A_{ij} \;=\; 1 \;. \tag{A230}$$

□

We will use the ∞-norm and the 1-norm of a matrix, which are defined based on the ∞-norm $\|\boldsymbol{x}\|_\infty = \max_i |x_i|$ and 1-norm $\|\boldsymbol{x}\|_1 = \sum_i |x_i|$ of a vector $\boldsymbol{x}$.

Definition A13. *The ∞-norm of a matrix is the maximum absolute row sum:*

$$\|\boldsymbol{A}\|_\infty \;=\; \max_{\|\boldsymbol{x}\|_\infty = 1} \|\boldsymbol{A}\,\boldsymbol{x}\|_\infty \;=\; \max_i \sum_j |A_{ij}| \;. \tag{A231}$$

The 1-norm of a matrix is the maximum absolute column sum:

$$\|\boldsymbol{A}\|_1 \;=\; \max_{\|\boldsymbol{x}\|_1 = 1} \|\boldsymbol{A}\,\boldsymbol{x}\|_1 \;=\; \max_j \sum_i |A_{ij}| \;. \tag{A232}$$

The statements of next theorem are known as Perron-Frobenius theorem for square stochastic matrices $\boldsymbol{A} \in \mathbb{R}^{n \times n}$, e.g. that the spectral radius $\mathcal{R}$ is $\mathcal{R}(\boldsymbol{A}) = 1$. We extend the theorem to a "∞-norm equals one" property for rectangular stochastic matrices $\boldsymbol{A} \in \mathbb{R}^{m \times n}$.

Lemma A5 (Perron-Frobenius)**.** *If $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ is a row-stochastic matrix, then*

$$\|\boldsymbol{A}\|_\infty \;=\; 1\,, \quad \left\|\boldsymbol{A}^T\right\|_1 \;=\; 1\,, \textit{ and for } n = m \quad \mathcal{R}(\boldsymbol{A}) \;=\; 1\,. \tag{A233}$$

Proof. $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ is a row-stochastic matrix, therefore $A_{ij} = |A_{ij}|$. Furthermore, the rows of $\boldsymbol{A}$ sum up to one. Thus, $\|\boldsymbol{A}\|_\infty = 1$. Since the column sums of $\boldsymbol{A}^T$ are the row sums of $\boldsymbol{A}$, it follows that $\left\|\boldsymbol{A}^T\right\|_1 = 1$.
For square stochastic matrices, that is $m = n$, Gelfand's Formula (1941) says that for any matrix norm $\|.\|$, for the spectral norm $\mathcal{R}(\boldsymbol{A})$ of a matrix $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ we obtain:

$$\mathcal{R}(\boldsymbol{A}) \;=\; \lim_{k \to \infty} \left\|\boldsymbol{A}^k\right\|^{1/k} \,. \tag{A234}$$

Since the product of row-stochastic matrices is a row-stochastic matrix, $\boldsymbol{A}^k$ is a row-stochastic matrix. Consequently $\left\|\boldsymbol{A}^k\right\|_\infty = 1$ and $\left\|\boldsymbol{A}^k\right\|_\infty^{1/k} = 1$. Therefore, the spectral norm $\mathcal{R}(\boldsymbol{A})$ of a row-stochastic matrix $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ is

$$\mathcal{R}(\boldsymbol{A}) \;=\; 1\,. \tag{A235}$$

The last statement follows from Perron-Frobenius theorem, which says that the spectral radius of $\boldsymbol{P}$ is 1. □

Using random matrix theory, we can guess how much the spectral radius of a rectangular matrix deviates from that of a square matrix. Let $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ be a matrix whose entries are independent copies of some random variable with zero mean, unit variance, and finite fourth moment. The Marchenko-Pastur quarter circular law for rectangular matrices says that for $n = m$ the maximal singular value is $2\sqrt{m}$ [79]. Asymptotically we have for the maximal singular value $s_{\max}(\boldsymbol{A}) \propto \sqrt{m} + \sqrt{n}$ [104]. A bound on the largest singular value is given by [122]:

$$s_{\max}^2(\boldsymbol{A}) \;\leqslant\; (\sqrt{m} \;+\; \sqrt{n})^2 \;+\; O(\sqrt{n}\,\log(n)) \text{ a.s.} \tag{A236}$$

Therefore, a rectangular matrix modifies the largest singular value by a factor of $a = 0.5(1 + \sqrt{n/m})$ compared to a $m \times m$ square matrix. In the case that tstates are time aware, transitions only occur from states s_t to s_{t+1}. The transition matrix from states s_t to s_{t+1} is denoted by $\boldsymbol{P}_t$.

States affected by the on-site variance $\boldsymbol{\omega}_k$ (reachable states). Typically, states in s_t have only few predecessor states in s_{t-1} compared to N_{t-1}, the number of possible states in s_{t-1}. Only for those states in s_{t-1} the transition probability to the state in s_t is larger than zero. That is, each $i \in s_{t+1}$ has only few $j \in s_t$ for which $p_t(i \mid j) > 0$. We now want to know how many states have increased variance due to $\boldsymbol{\omega}_k$, that is how many states are affected by $\boldsymbol{\omega}_k$. In a general setting, we assume random connections.
Let N_t be the number of all states s_t that are reachable after t time steps of an episode. $\bar{N} = 1/k \sum_{t=1}^k N_t$ is the arithmetic mean of N_t. Let c_t be the average connectivity of a state in s_t to states in s_{t-1} and $\bar{c} = \left(\prod_{t=1}^k c_t\right)^{1/k}$ the geometric mean of the c_t. Let n_t be the number of states in s_t that are affected by the on-site variance $\boldsymbol{\omega}_k$ at time k for $t \leqslant k$. The number of states affected by $\boldsymbol{\omega}_k$ is $a_k = \sum_{t=0}^k n_t$. We assume that $\boldsymbol{\omega}_k$ only has one component larger than zero, that is, only one state at time $t = k$ is affected: $n_k = 1$. The number of affected edges from s_t to s_{t-1} is $c_t n_t$. However, states in s_{t-1} may be affected multiple times by different affected states in s_t. Figure A1 shows examples of how affected states affect states in a previous time step. The left panel shows no overlap since affected states in s_{t-1} connect only to one affected state in s_t. The right panel shows some overlap since affected states in s_{t-1} connect to multiple affected states in s_t.
The next theorem states that the on-site variance $\boldsymbol{\omega}_k$ can have large effect on the variance of each previous state-action pair. Furthermore, for small k the number of affected states grows exponentially, while for large k it grows only linearly after some time $\hat{t}$. Figure A2 shows the function which determines how much a_k grows with k.

Theorem A10. *For $t \leqslant k$, $\boldsymbol{\omega}_k$ contributes to $\boldsymbol{V}_t^\pi$ by the term $\boldsymbol{P}_{t \leftarrow k}\, \boldsymbol{\omega}_k$, where $\|\boldsymbol{P}_{t \leftarrow k}\|_\infty = 1$. The number a_k of states affected by the on-site variance $\boldsymbol{\omega}_k$ is*

$$a_k \;=\; \sum_{t=0}^k \left(1 \;-\; \left(1 \;-\; \frac{c_t}{N_{t-1}}\right)^{n_t}\right) \, N_{t-1}\,. \tag{A237}$$